CEST perspectives for human brain applications at 11.7 T

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Synopsis

Motivation: High magnetic field CEST imaging presents unique opportunities for clinical applications development.

Goal(s): To investigate the potential of human brain CEST imaging on the 11.7 T scanner using tailored parallel transmission CEST pulses and compare performances with those of 7 T.

Approach: 7 T and 11.7 T B_1^+ maps were acquired on a realistic human head phantom and pTx CEST pulses were designed for a targeted B_1^+ value under different energy management strategies.

Results: At 11.7 T, the optimized CEST pulses exhibit more inhomogeneity than at 7 T. pTx and VOPs are highly beneficial, with best homogeneity obtained for hydroxyl-weighted maps.

Impact: This study explores the potential of using the 11.7 T scanner for human brain CEST applications, addressing B_1^+ inhomogeneities and SAR compliance. It marks a first step toward clinical evaluation of CEST at magnetic fields higher than 7 T.

Introduction

High magnetic fields significantly improve the Chemical Exchange Saturation Transfer (CEST) contrast, enhancing the detection sensitivity and the spectral resolution. High field CEST imaging has already shown encouraging pre-clinical and clinical applications for brain tumor characterization with 9.4 T creatine CEST[1] or non-invasive visualization of seizure foci with 7 T glutamate CEST[2]. In 2024, a groundbreaking 11.7 T whole-body scanner, "Iseult", was launched[3], promising to push the limits of MRI applications including CEST. Working at high magnetic field comes however with its challenges such as enhanced B_0 and B_1^+ inhomogeneities and increased specific absorption rate (SAR) levels. A prior study[4] suggested combining parallel transmission (pTx) with virtual observation points[5] (VOPs) to reduce B_1^+ field inhomogeneities, and at the same time, ensuring compliance with SAR limits. In this study, we aim to investigate the potential of clinical CEST imaging on the 11.7 T scanner using a similar approach and compare its performance with that of 7 T.

Methods

B_1^+ maps acquisitions

B_1^+ maps were acquired on the 11.7 T and 7 T MR systems using an in-house head coil[6] and a Nova head coil, respectively, both in 8Tx-32Rx configuration. Acquisitions were performed on a realistic 3D-printed human head phantom (filled with distilled water, 0.064 mM Dotarem and 4 g/L NaCl to mimic brain tissue properties) using an interferometric turbo-FLASH sequence[7].

CEST pulses optimization

Tailored pTx CEST pulses, consisting of two modes (repeated N times), were designed to achieve whole-brain or slab selective saturation (5 mm thickness, axial and isocentered) by minimizing the L2-norm between the effective $B_{1,RMS}^+$ and a given target[4,8].

The optimization was performed under hardware (peak and total power) and SAR constraints, using either VOPs or manufacturer limits (protected mode, PM). VOPs allow real-time SAR supervision, which has been shown to improve the homogeneity of the CEST saturation pulses[4]. Under VOP supervision, peak channel and total power were limited to 3 W and 16 W, respectively, representing twice the limits allowed in PM. Local SAR for 11.7 T and 7 T optimization was limited to 20 W/kg ("Level 1") and 10 W/kg ("Normal" mode), respectively. Global SAR for both scanners was limited to 3.2 W/kg.

To evaluate the performance of the calculated pulses, we computed the root mean square error (RMSE) of the resulting $B_{1,RMS}^+$ and represented it as a function of $B_{1,RMS}^+\sqrt{(DC)}$, where $DC = T_{sat}/TR$ is the duty cycle, and T_{sat} and TR are the CEST saturation time and the repetition time, respectively. For comparison, we also simulated the performance of the circular polarized mode. $B_{1,RMS}^+$ maps were simulated with the optimal found pulses for typical acquisitions parameters for glutamate-, hydroxyl- and APT-weighted CEST[9]. Parameters used are listed in Table 1.

Such optimized CEST pulses will be used in a custom-made 2D fast steady-state CEST sequence[10] which has been implemented on the 11.7 T scanner and is currently being tested.

Results and discussion

7 T simulations (Fig. 1) align with previously reported findings[4], demonstrating that tailored pulses using VOPs achieve the best performance for both whole-brain and slab applications: ~ 1.6% RMSE on a slab when $B_{1,RMS}^+\sqrt{(DC)}$≤1 µT.

Whole-brain or slab 11.7 T pulses exhibit more inhomogeneity on average compared to 7 T pulses (Fig. 1). As expected, focusing on a slab reduces the RMSE (Fig. 1), with up to a 70% reduction in RMSE on a slab compared to the whole brain when using tailored pulses with VOPs.

Simulations further suggest that VOPs will be essential for 11.7 T CEST applications, as for tailored pulse optimization on a slab, using VOPs results in a 35% average RMSE improvement compared to PM optimization.

In the context of future CEST applications, Fig. 2 displays $B_{1,RMS}^+$ maps obtained with optimal found pulses for glutamate-, hydroxyl-, and APT-weighted CEST on a slab at 11.7 T and 7 T. Despite optimization, residual inhomogeneity is still visible on 11.7 T maps contrary to 7 T maps. Best 11.7 T homogeneity is obtained for hydroxyl-weighted maps (Fig. 2A) with RMSE = 10%. As shown in Fig. 2E, increasing saturation power, as needed for glutamate, constrains optimization due to local SAR limits, leading to a higher RMSE value of 16%. As an example, optimized pulses for APT-weighted CEST are shown in Fig. 3.

Conclusion

This work highlights the importance of using pTx tailored pulses with VOPs for 11.7 T CEST applications. The results are promising, especially for low-saturation CEST pulses, however, addressing the residual B_1^+ inhomogeneity remains essential. Future solutions may involve novel RF coil designs[11] possibly with additional transmitter channels and the implementation of B_1^+ post-processing corrections[12].

Acknowledgements

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Figures

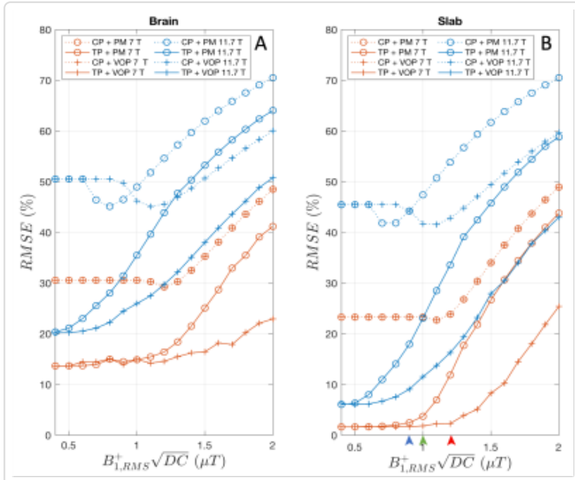

Figure 1: $B_{1,RMS}^+$ root mean square error (RMSE) of circular polarized (CP) and tailored pulses (TP) for 7 T and 11.7 T CEST optimization under different energy management strategies (PM or VOPs). Arrows point to the value of $B_{1,RMS}^+\sqrt{(DC)}$ used for saturation pulses for hydroxyl-weighted (blue), APT-weighted (green) and glutamate-weighted (red) CEST. Optimizations are performed on a phantom.

	B_1^+ target (µT)	TR (s)	T_{sat} (s)
Hydroxyl-weighted CEST	1.5	2.5	1
APT-weighted CEST	2	4	1
Glutamate-weighted CEST	3	5	0.8

Table 1: Parameters used for tailored pulse design of CEST pulses for hydroxyl-, APT-, and glutamate-weighted CEST.

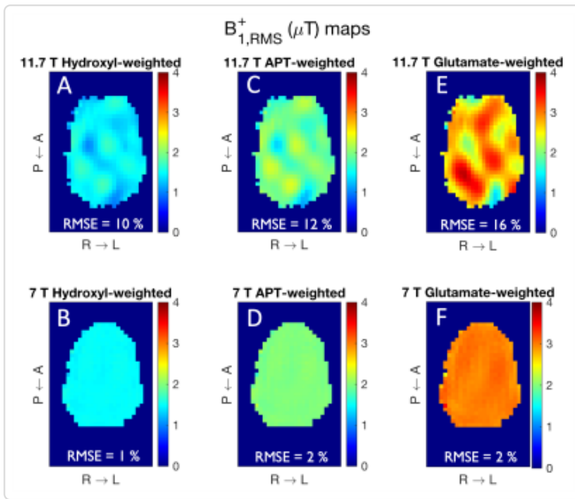

Figure 2: Simulated $B_{1,RMS}^+$ maps obtained with optimized tailored pulses for CEST saturation at 11.7 T and 7 T, for hydroxyl-weighted (A,B), APT-weighted (C,D) and glutamate-weighted (E,F) CEST. Simulations are performed on an axial, isocentered slab of 5 mm thickness.

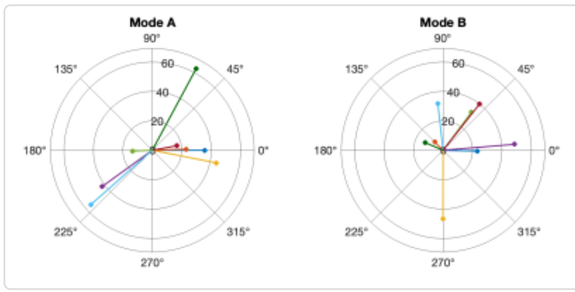

Figure 3: Magnitude and phase of the optimized tailored pulse for 11.7 T APT-weighted CEST. The pulse consists of two Gaussian-Hanning modes of 50 ms duration each. Increasing the number of modes did not improve optimization. Each color represents an RF channel, with bar length representing the amplitude (V).